

K2-30b

R-2346

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_210219 · created 2026-08-19T00:49:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459264.6828 +/- 0.0084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.045
Transit depth (Rp/Rs)^2	1.54 +/- 1.12 [%]
Orbital Inclination (inc)	87.7 +/- 2.21
Airmass coefficient 1 (a1)	1.034 +/- 0.034
Airmass coefficient 2 (a2)	-0.0115 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	3.26 %
Best Comparison Star	#3 - [278, 37]
Optimal Method	PSF photometry
Transit Duration (day)	0.089 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.124 $\pm$ 0.045 vs 0.1266 (deviation 0.06 σ)
Duration fit vs published	0.089 d vs 0.0971 d (deviation 0.28 σ)
Mid-time O-C	-9.5 min
Depth SNR	2.8
Residual scatter	3.26 %

Light curve

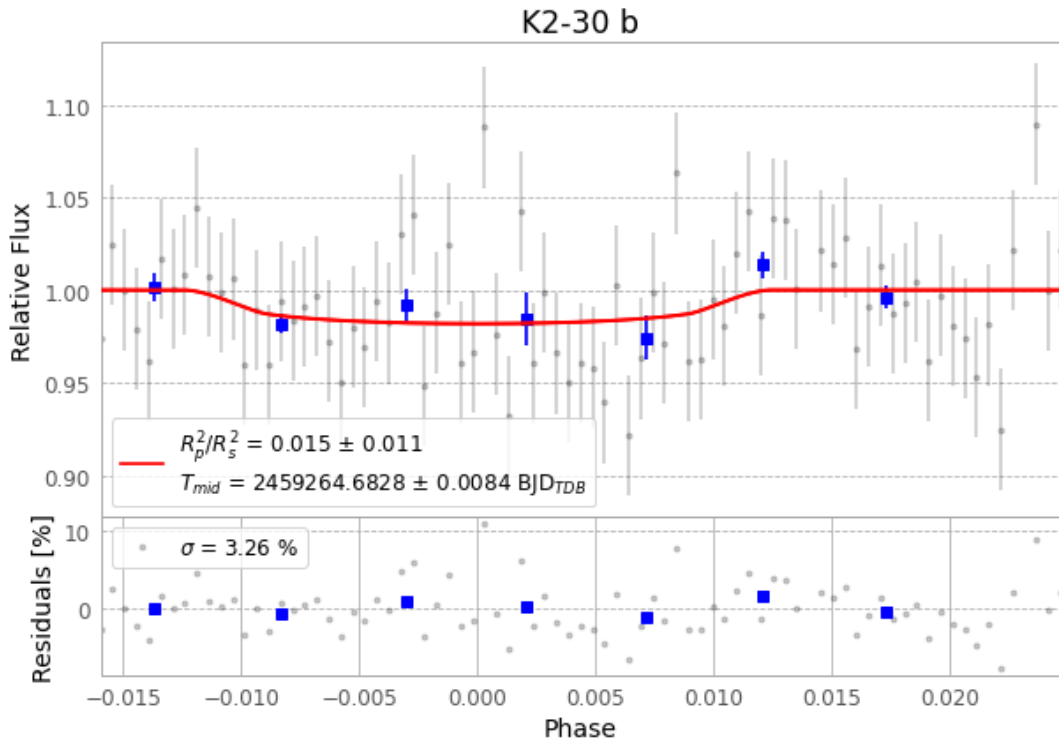


Figure 1 — FinalLightCurve_K2-30 b_2021-02-18.png (SHA-256 2ad1ee5596431c28...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 228], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a1554f915c153eb2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

85 FITS frames (56 MB), K2-30210219024812.FITS → K2-30210219070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-210219.log (13ac434b497a...), FinalLightCurve_K2-30 b_2021-02-18.png (2ad1ee559643...), FinalLightCurve_K2-30 b_2021-02-18.csv (160c7336187a...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:49Z.